Posting Like a State: How do Citizens Interact with Autocrats' Media

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Abstract

How should scholars interpret engagement with official media in authoritarian settings? Existing accounts often read digital traces on state platforms either as evidence of propaganda success or as noise generated by skeptical audiences. I argue instead that official accounts are mixed channels that bundle public service information with political messaging, and that citizens sort their responses according to the visibility of the expressive act. Using 4.21 million post titles from 329 Chinese local-government WeChat accounts, the paper combines a title-based classification workflow with platform-centered measurement of reads, likes, shares, and *zaikan*. The empirical strategy focuses on how interface reforms changed the meaning and visibility of engagement, with particular attention to the distinction between low-visibility approval and socially legible endorsement. This framework recasts engagement metrics as strategic behavioral traces rather than direct measures of regime support and extends preference falsification theory to layered digital expression.

Keywords: digital authoritarianism; propaganda; official social media; WeChat; preference falsification; state–society interaction.

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1 Introduction

How do citizens in autocracies interact with official social media, and what do those traces actually mean? As propaganda, public communication, and service delivery migrate onto digital platforms, state-controlled accounts accumulate vast numbers of reads, likes, and shares. These metrics are tempting to interpret as indicators of propaganda success or regime support. Yet in authoritarian settings, interacting with official content is rarely a purely expressive act. Citizens may engage because they need information, because a post is difficult to avoid, because approval is low-cost, or because visible endorsement carries reputational and political risk. The political meaning of digital engagement is therefore fundamentally ambiguous.

Existing work speaks to pieces of this puzzle, but rarely to the mechanism at stake here. Research on Chinese propaganda shows that the party-state competes for attention through commercialization, soft propaganda, distraction, and popularization rather than relying on crude indoctrination alone (Brady 2008; King, Pan, and Roberts 2017; Lu and Pan 2021; Mattingly and Yao 2022; Repnikova and Fang 2018; Roberts 2018; Stockmann 2013; Wang 2024; Zou 2021). Research on preference falsification and self-censorship emphasizes the gap between private belief and public expression under pressure (Carter, Carter, and Schick 2024; Jiang and Yang 2016; Kuran 1995; Noelle-Neumann 1993; Shen and Truex 2020). A third literature shows that government WeChat accounts are not merely propaganda outlets: local authorities also use them for public service delivery, routine contact, and audience acquisition (Jiang et al. 2021; Liu, Ma, and Xia 2024; Yu and Shen 2026; Zheng 2013; Zheng and Zheng 2014). A related strand highlights semi-official accounts, delegated censorship, and layered platform governance, suggesting that state influence is often embedded in organizationally ambiguous or platform-native forms (Neagli 2021; Sun and Zhao 2022; Tai and Fu 2020). What these literatures do not yet explain well is how platform design changes the political meaning of interaction itself.

I argue that official social media in digital authoritarianism are mixed channels that bundle public service information with political messaging, and that citizens sort their behavior across different levels of expressive risk. On WeChat, reading is low-risk consumption, a like is low-

visibility approval, *zaikan* is socially legible endorsement, and sharing is the highest-visibility form of amplification. This layered-expression perspective shifts the theoretical question away from whether citizens "believe" propaganda and toward how they manage observable traces of alignment.

I evaluate this argument using 4.21 million post titles published by 329 prefecture-level government WeChat official accounts between 2015 and 2024. The article makes two empirical moves. First, I use a theory-driven 'LLM + human' classification workflow to assign fine-grained post labels and then collapse them into three main content families: hard propaganda, soft propaganda, and public service. Second, I use the 2018 and 2020 WeChat interface reforms to motivate a redesigned empirical analysis of how platform changes alter the meaning of a click.

This article makes three contributions. First, it extends preference falsification theory from speech to platform-mediated behavior by showing that digital expression is layered across reading, liking, *zaikan*, and sharing. Second, it contributes to the study of propaganda and media consumption by separating attention from endorsement and by showing how service content both builds and competes for political attention. Third, it recasts official social media as an everyday site of authoritarian state–society interaction, where governments manage not only information but also the terms under which citizens can safely engage.

2 Theory and Hypotheses

Official social media in authoritarian settings should be understood as a platformed interaction regime rather than as a simple channel of state persuasion. In classic preference falsification accounts, the central distinction is between private preference and public statement. On platforms such as WeChat, however, expression is not binary. The interface decomposes response into multiple actions that expose users to different audiences and therefore to different expressive risks. The state may observe aggregate traces, but friends, colleagues, and acquaintances can also infer meaning from visible acts. Digital preference falsification is therefore layered: the same user may

privately consume official content, register low-visibility approval, and avoid socially visible endorsement (Kuran 1995; Noelle-Neumann 1993; Shen and Truex 2020; Tsai and Men 2018; Xu 2022).

This layered-expression logic matters because Chinese government WeChat accounts are mixed channels rather than purely political outlets. Existing research shows that state agencies use social media not only to disseminate political messages but also to provide services, route transactions, manage crises, and sustain routine interaction with the public (Jiang et al. 2021; Liu, Ma, and Xia 2024; Yu and Shen 2026; Zheng 2013; Zheng and Zheng 2014). That literature is especially useful because it shows that many users approach official accounts as lookup tools or service interfaces rather than as ideological venues. Service content therefore helps governments build durable audiences, normalize repeated contact, and increase the probability that users encounter future political content. At the moment of consumption, however, service information also competes with overt political messaging for scarce attention. This is the paper's attention dilemma.

The same logic also changes how we should think about propaganda. Contemporary influence online often works through participatory persuasion, semi-official intermediaries, and delegated forms of control rather than through a single visible propaganda apparatus (King, Pan, and Roberts 2017; Lu and Pan 2021; Neagli 2021; Repnikova and Fang 2018; Sun and Zhao 2022). Hard propaganda foregrounds hierarchy, ideology, discipline, and coercive state capacity. Soft propaganda packages regime-favorable messages in more appealing, emotionally resonant, or developmental frames (Mattingly and Yao 2022; Pan, Shao, and Xu 2022; Wang 2024). Citizens may read all three kinds of content, but the willingness to leave an observable trace should vary with both content risk and interface design. Platform reforms do not merely change aesthetics. They reprice the political and social costs of interacting with the state.

This framework yields four expectations:

H1 (Attention and Utility). Public service content should attract more routine consumption than either hard or soft propaganda because it provides immediate utility and lower expressive stakes.

H2 (Rebundling Visible Approval). When the 2018 redesign bundles approval with public rec-

ommendation, low-visibility like responses should collapse and observed reactions should be reallocated into more socially legible channels.

H3 (Private Substitution). When low-visibility likes return in 2020, approval should shift back toward likes rather than rising uniformly across all public-facing reactions.

H4 (Political-Risk Gradient). These reallocations should be strongest for hard propaganda, then soft propaganda, and weakest for routine public-service content.

Table 1: Platform reforms and theoretical expectations

Time point	Brief policy content	Reading / exposure	Visible endorsement	Low-risk approval
2017–05 (v6.5.8 rollout)	WeChat introduces "See" (*kanyikan*), adding a friend- and algorithm-based discovery channel while leaving the bottom "like" button unchanged.	+	0	0
2018–12–21	WeChat 7.0 replaces "like" with "good-looking" (*haokan*, later *zaikan*), bundling approval with recommendation into "See" (*kanyikan*).	0	reallocated / ambiguous	–
2020–06–29	The "like" button is restored at the bottom of public-account articles alongside the existing "seen" (*zaikan*) and share buttons.	0	0/–	+

Note: '+' indicates an expected increase, '-' an expected decrease, and '0' little immediate change. The 2018 redesign mechanically rebundles approval with recommendation, so observed public-signal measures may be reallocated rather than move with a single clean sign. The strongest directional effects are expected for hard propaganda.

3 Setting, Data, and Measurement

WeChat is the central digital infrastructure of everyday life in China. Since its launch in 2011, it has evolved into a multipurpose "super app" that combines messaging, payments, news distribution, and social networking within a single platform. By the 2020s, it had become the primary interface through which Chinese citizens communicate, consume news, and access digital public and private services. For a study of official digital communication in China, WeChat matters not because it is just another social media platform, but because it is deeply embedded in ordinary social, administrative, and economic life (Ng 2015; Roberts 2020).

Within this ecosystem, "public accounts" operate as mixed institutional channels. They transmit propaganda and political messaging, publicize leadership priorities, and provide highly practical information and service announcements (Jiang et al. 2021; Liu, Ma, and Xia 2024; Yu and Shen 2026; Zheng 2013; Zheng and Zheng 2014). At the same time, the broader WeChat ecosystem includes semi-official and organizationally ambiguous accounts that can carry state-aligned messages without looking fully official (Neagli 2021). A typical article displays a read counter together with several engagement mechanisms: a like button, *haokan*/*zaikan*, sharing, and collecting. These buttons do not measure the same thing with different labels. They ask users to leave different kinds of traces under different conditions of visibility.

The key institutional changes are the 2018 and 2020 interface reforms. In December 2018, WeChat replaced the original like button on newly pushed public-account articles with a "good-looking" button tightly integrated with the "See" tab. A previously quasi-private like was transformed into a socially legible recommendation. In June 2020, WeChat restored the traditional like button alongside *zaikan* and sharing, thereby unbundling low-visibility approval from public recommendation. I treat 2017 as background discovery reform and 2018/2020 as the two main discontinuities for expressive engagement. Table 1 summarizes these shifts.

This study uses the universe of posts published by 329 prefecture-level government WeChat public accounts between 1 January 2015 and 31 December 2024, covering 31 provincial-level units and 4,207,072 posts. The unit of observation for classification is the individual post title. For each post, I record the account identifier, publication date, title, and engagement indicators displayed in the interface, including reads, likes, shares, collecting, and *haokan*/*zaikan*. The archived source file also preserves a first-pass category field, a short reason field, a keyword field, and a confidence score from an earlier coarse classification pass; these metadata are used only for title-based normalization and coding, not as outcomes. Because WeChat top-codes some counters at 100,000 (displayed as "100k+"), I recode those values to 100,001 so they remain ordered while preserving censoring at the upper bound.

To categorize post content at scale, I use an LLM as a text coder rather than as a generator.

The production prompt is set before full-corpus assignment and uses only title-side inputs: title, account name, province, city, publication date, keywords, and the short reason field. Engagement outcomes, account-level outcome patterns, and reform-period indicators never enter the classification prompt. The coding workflow is built on a fixed ten-category taxonomy. The main text relies on this workflow only for measurement credibility; the SI documents taxonomy construction, coder-audit rules, and validation diagnostics.

I validate the LLM labels against a dual-coded gold-standard sample of 2,000 post titles coded under the same codebook and adjudicated to a final reference label. The SI reports the reliability protocol, aggregate validation metrics, class-wise performance, and a family-level confusion matrix. For the main analyses, I collapse the fine-grained labels into three theoretically motivated families: hard propaganda, soft propaganda, and public service. Hard propaganda includes overtly political and disciplinary content; soft propaganda packages regime-favorable messages through development, culture, tourism, and performance narratives; public service covers instrumental governance content such as procedures, welfare, health, traffic, weather, and emergency guidance. State-governance categories remain in the data for balance checks and appendix material but are not the paper's main theoretical families.

The paper's measurement logic is therefore theory-driven rather than purely unsupervised. The LLM solves the assignment problem at the post level; it does not decide which families matter theoretically. That distinction is important because the theoretical categories come from prior literature on mixed governance communication, soft propaganda, and expressive visibility rather than from an inductive topic model alone.

4 Empirical Analysis

The descriptive analysis, identification strategy, and causal results are being rebuilt from the cleaned 2015–2024 title-level dataset and the revised measurement workflow documented in the appendix. Subsequent empirical sections will report new descriptive patterns, redesigned discontinuity anal-

yses, and updated heterogeneity tests using the current classification baseline rather than the superseded specification.

5 Conclusion

Official engagement metrics on authoritarian platforms are not transparent measures of support. They are shaped by mixed content environments and by the visibility structure of the interface itself. On WeChat, the central analytical task is therefore to separate attention from endorsement and to study how interface design conditions the forms of engagement that users are willing to leave behind.

The broader implication is that digital authoritarian control works partly by structuring traces, not only by changing beliefs. Once we distinguish low-risk consumption from low-visibility approval and from socially legible endorsement, the political meaning of a click changes. That insight extends preference falsification theory to platform-mediated behavior and suggests a more cautious reading of engagement metrics in authoritarian settings.

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Online Supplemental Information

A WeChat, Public Accounts, and Interface Reforms **SI-1**

A.1 Platform Scale and Official-Account Infrastructure SI-1

A.2 Why Public Accounts Matter for Local Government SI-1

A.3 Interface Reforms Relevant to Engagement Measures SI-1

B WeChat Title Data and Classification Validation **SI-2**

B.1 Taxonomy Construction and Discovery Review SI-2

B.2 Human Coding Rules and Validation Sample SI-3

B.3 LLM Prompt SI-3

B.4 Validation Results SI-5

A WeChat, Public Accounts, and Interface Reforms

A.1 Platform Scale and Official-Account Infrastructure

Weixin and its international sister product WeChat are not merely messaging applications. They are a broad digital infrastructure that combines chat, social networking, public content distribution, payments, mini programs, and search within the same app environment. Tencent's 2024 annual results report 1.385 billion combined monthly active accounts as of 31 December 2024, while Tencent's product materials describe the platform as an ecosystem spanning Moments, Official Accounts, Weixin Pay, Mini Programs, Search, Channels, and related services (Tencent 2026b; Tencent Holdings Limited 2025). For this paper, the most important layer is the official-account system. Tencent's own description of Weixin Official Accounts presents it as a publishing and notification infrastructure through which individuals and organizations gather followers, push content, and send service or marketing notifications. The same page reports more than 20 million official accounts as of February 2021 and distinguishes between subscription accounts, oriented toward content publication and readership management, and service accounts, oriented toward user management and customer service (Tencent 2026a).

A.2 Why Public Accounts Matter for Local Government

This architecture matters for local-government communication because official accounts are embedded inside ordinary social and administrative life rather than separated into a standalone news app. A user may encounter a government article through a subscription feed, a chat share, a Moments post, a search result, or a recommendation channel such as *kanyikan* ("See"). Once opened, the article sits inside the same environment that also contains private chats, group chats, payments, and service tools. That is why official accounts can function at once as broadcast channels, low-friction service interfaces, and distribution nodes inside the larger WeChat ecosystem. Tencent's 2017 first-quarter results, for example, already emphasized that official-account content was being tied more closely to search and mini-program traffic through embedded links and discovery tools (Tencent Holdings Limited 2017). Existing research on government WeChat accounts reaches a similar conclusion from the user side: many citizens treat these accounts as routine lookup and service tools rather than as purely political venues (Jiang et al. 2021; Liu, Ma, and Xia 2024; Yu and Shen 2026). For the purposes of this paper, the key implication is that article-level interaction takes place inside a platform where private communication, semi-public recommendation, and public-facing service use are tightly bundled together.

A.3 Interface Reforms Relevant to Engagement Measures

The article-level engagement measures used in this paper were shaped by a small number of platform redesigns that changed what users could do from the bottom of a public-account article page. First, contemporary reporting from late July 2014 documented that WeChat began displaying article-level read counts and a visible like button on public-account pages, creating the first widely visible article-side counters (People.cn 2014). Second, in May 2017, WeChat introduced "Search" and "See" through WeChat Lab and the discovery tab, broadening how users could encounter official-account content beyond simple subscription order (Sina Tech 2017; Tencent Hold-

Table SI.1: Final ten-label taxonomy, higher-order families, and corpus shares.

detailed_label	family	n_posts	share_posts
Ideological Education	Hard Propaganda	470,939	0.112
Leadership and Political Events	Hard Propaganda	530,382	0.126
Public Service Information	Public Service	524,185	0.125
Social Protection and Welfare	Public Service	274,601	0.065
Emergency and Risk Communication	Public Service	544,084	0.129
Policy and Administrative Disclosure	State Governance	432,057	0.103
Social Governance and Enforcement	State Governance	203,340	0.048
Mass Mobilization and Participation	State Governance	96,840	0.023
Economic Development and Construction	Soft Propaganda	641,160	0.152
City Image and Cultural Activity	Soft Propaganda	489,484	0.116

ings Limited 2017). Third, on 21 December 2018, WeChat 7.0 replaced the older like button with *haokan* ("good-looking"), and press coverage at the time emphasized that clicking *haokan* could sync the article into the "See" feed and make the signal visible to friends (Zhoudao 2018). Fourth, on 15 March 2019, *haokan* was renamed *zaikan* (roughly, "seen" or "reading now"); coverage framed the shift as softer wording for the same socially visible recommendation channel rather than as a new interaction type (TechNode 2019). Finally, on 29 June 2020, the article footer was redesigned again: like returned alongside *zaikan*, and a dedicated share button was added, making it easier to send articles to friends or Moments while also restoring a lower-visibility approval channel (Securities Times 2020).

Taken together, these reforms changed not only how official-account content circulated but also the social meaning of leaving a trace on an article page. In this paper's design, 2017 therefore serves mainly as a discovery and distribution reform, 2018 as a visibility-regime change in one-click approval, 2019 as a useful rename placebo, and late June 2020 as the return of a distinct lower-visibility like channel.

B WeChat Title Data and Classification Validation

This section documents the title-based WeChat classification pipeline used in the main analyses. The cleaned corpus covers all posts published by 329 prefecture-level government WeChat public accounts between 1 January 2015 and 31 December 2024. The key measurement constraint is that the archived source file preserves post titles consistently across the full nationwide collection but does not preserve article bodies in the current working archive. The unit of text classification is therefore the individual post title. In addition to the title itself, the working archive preserves auxiliary text-side cues such as a short keyword field, a short reason field, and a confidence score. These metadata are used only as classification inputs. Engagement outcomes such as reads, likes, shares, and *zaikan* never enter the classification prompt.

B.1 Taxonomy Construction and Discovery Review

The final taxonomy fixes ten mutually exclusive detailed labels and four higher-order families. Table SI.1 reports the final taxonomy, the number of posts assigned to each label, and each label's share of the cleaned corpus.

The codebook was finalized after a 400-title discovery review carried out with experienced media practitioners to surface both common labels and difficult boundary cases. As Table SI.2

Table SI.2: Discovery-sample design used to refine category boundaries and codebook rules.

phase	target	rule
Round 1	200	Balanced grid across five time bins and the ten cleaned labels
Round 2A	50	Short or low-information titles
Round 2B	50	Low-confidence titles
Round 2C	50	Mixed-domain titles spanning multiple coding cues
Round 2D	50	Adjacent-boundary titles from semantically neighboring categories

shows, the first round balanced across five time bins and the ten cleaned labels, while the second round concentrated on short or low-information titles, low-confidence titles, mixed-domain titles, and adjacent boundary cases.

That discovery review clarified the main inclusion and exclusion rules that later structure both human coding and model prompting. The hardest boundaries are not random. They recur where title wording mixes leadership reporting with development claims, routine service information with welfare substance, policy release with enforcement, emergency warning with ordinary governance control, and development performance with city branding. Table SI.3 summarizes the resulting coding cues and the corresponding title patterns.

That discovery review clarified the boundary rules later used in both human coding and model prompting. Its role was to stabilize the codebook and the main-communicative-purpose rule before full-corpus assignment rather than to supply any supervised training set in the machine-learning sense.

B.2 Human Coding Rules and Validation Sample

The validation audit drew 2,000 titles with sampling probabilities proportional to city-level corpus size, while retaining at least one title from every city in the corpus. This preserves exposure to the largest posting environments without collapsing the validation sample into only a handful of high-volume cities. Table SI.4 summarizes the resulting audit sample.

Two independent coders then assigned each title one and only one primary label and the corresponding higher-order family. The operative rule was to code the title's *main communicative purpose*, not the broadest noun phrase or the most politically salient word in isolation. In practice, this meant preferring public-service information over welfare only when the title mainly explained how to do something; preferring leadership and political events over policy disclosure when leaders, meetings, inspections, speeches, or appointments were the center of the title; separating policy disclosure from governance and enforcement by distinguishing rule release from rule enforcement; and separating development performance from city-image content by distinguishing projects and economic upgrading from tourism, festivals, and branding. Coders were blind to model output and to all engagement outcomes. Disagreements were adjudicated to produce the final gold-standard label. The 2,000-title audit sample was held out from prompt design and used only for validation.

B.3 LLM Prompt

The full-corpus pass used OpenAI with `model = gpt-5.4` and `reasoning_effort = low` under a deterministic single-label instruction. The admissible Chinese primary labels were 意识形态与宣传教育，时政与领导活动，公共服务信息，社会保障与公共福利，应急管理与风险沟通，政策与政务公开，社会治理与执法通报，群众动员与社会参与，经济与发展建设，

Table SI.3: Illustrative coding cues and title patterns for the ten detailed labels.

detailed_label	family	decision_cue	example_title_pattern
Ideological Education	Hard Propaganda	Political learning, Party education, discipline, and value transmission	"Study and implement the Party Congress spirit" style political-study or Party-education title
Leadership and Political Events	Hard Propaganda	Leaders, meetings, inspections, speeches, appointments, and deployments	"Municipal Party secretary inspects major projects" style leader-activity or meeting report
Public Service Information	Public Service	Practical guidance, reminders, procedures, notices, and daily-use service information	"Housing provident fund application guide" style practical procedure or reminder title
Social Protection and Welfare	Public Service	Education, health, employment, subsidies, pensions, and welfare provision	"Medical reimbursement policy adjustment" style welfare, education, health, or subsidy title
Emergency and Risk Communication	Public Service	Warnings, hazards, disasters, epidemics, safety alerts, and urgent risk communication	"Blue rainstorm warning" style safety warning, hazard, or epidemic-risk title
Policy and Administrative Disclosure	State Governance	Policy texts, formal notices, administrative disclosure, budgets, and statistics	"2024 government work report" style policy-release or administrative-disclosure title
Social Governance and Enforcement	State Governance	Enforcement, punishment, inspections, anti-fraud, and order-maintenance action	"Special rectification campaign bulletin" style enforcement or rectification title
Mass Mobilization and Participation	State Governance	Calls for participation, volunteering, campaigns, voting, and civic mobilization	"Civilized city creation initiative" style participation or volunteering title
Economic Development and Construction	Soft Propaganda	Projects, investment, industry, infrastructure, agriculture, and development performance	"Major projects break ground" style development-performance or infrastructure title
City Image and Cultural Activity	Soft Propaganda	Tourism, culture, festivals, exhibitions, place branding, and local storytelling	"Cherry blossom festival opens" style culture, tourism, festival, or branding title

Table SI.4: Coverage summary for the 2,000-title validation sample.

metric	value
Annotated titles	2,000
Cities represented	329
Provincial-level units represented	31
Minimum titles per city	1
Median titles per city	5
Maximum titles per city	25

Table SI.5: Inter-coder reliability for the dual-coded 2,000-title validation sample.

statistic	value
Primary-label exact agreement	0.8875
Family-label exact agreement	0.9130
Primary-label Cohen's kappa	0.8726
Family-label Cohen's kappa	0.8811
Adjudication rate	0.1125

and 城市形象与文化活动. The production prompt was set before large-scale assignment and used for both the corpus pass and the validation audit. The prompt was:

Prompt used for title-based coding

```
You are a Chinese political-communication annotator.
Task: classify a local-government WeChat post into a fixed ten-label taxonomy.
Use only title-side inputs. Do not use engagement metrics or infer category from popularity.
Choose exactly one primary label according to the post's main communicative purpose.
Decision rule: ask whether the title mainly teaches political values, reports leaders or political
events, provides practical service information, describes welfare or benefit provision, manages
immediate risk, discloses policy or administrative procedure, reports enforcement or governance action,
mobilizes participation, promotes development, or projects culture and city image.
Boundary rules:
- public service information vs. social protection and welfare: procedure and daily-use guidance vs.
welfare substance, health, education, jobs, subsidies, and pensions;
- leadership and political events vs. policy disclosure: leaders, meetings, inspections, speeches, and
appointments vs. policy texts, notices, budgets, and formal disclosure;
- policy disclosure vs. governance and enforcement: rule release vs. rule enforcement;
- emergency and risk communication vs. governance and enforcement: urgent warnings and hazards vs.
routine control and punishment;
- economic development vs. city image: projects and performance claims vs. tourism, festivals,
exhibitions, and branding.
Return JSON only in the following schema:
{"primary_label":"...", "family_label":"...", "confidence":0.00, "brief_rationale":"..."}
Input fields:
title = {title}
public_account_name = {public_account_name}
province = {province}
city = {city}
publish_date = {publish_date}
keywords = {keywords}
reason = {reason}
```

There was no fine-tuning stage and no supervised training split in the machine-learning sense. The discovery sample was used to finalize the codebook and prompt, after which the 2,000-title coder audit was held out for validation.

B.4 Validation Results

Table SI.5 reports the reliability of the dual-coded audit sample. Exact agreement reaches 0.8875 at the detailed-label level and 0.9130 at the family level. Cohen's κ is 0.8726 for the ten-label task and 0.8811 for the four-family task. Only 11.25% of titles required adjudication, which is consistent with the fact that the hardest disagreements cluster around a small set of expected boundary zones rather than around wholesale coder confusion.

Against the adjudicated gold-standard labels, the LLM reaches 0.8150 accuracy and 0.7955 macro F1 on the ten-label task. At the family level, accuracy rises to 0.8755 and macro F1 to 0.8642. Table SI.6 reports both sets of overall validation metrics together. This higher family-level

Table SI.6: Overall validation metrics for detailed-label and family-level LLM coding.

task	accuracy	macro_f1	evaluation_titles
Detailed labels	0.8150	0.7955	2,000
Higher-order families	0.8755	0.8642	2,000

Table SI.7: Class-wise validation metrics for the four higher-order families.

family	precision	recall	f1	support
Hard Propaganda	0.8478	0.8966	0.8715	435
Public Service	0.8959	0.9016	0.8987	630
Soft Propaganda	0.9455	0.8691	0.9057	619
State Governance	0.7567	0.8070	0.7810	316

performance is substantively important because the paper's theory-facing analyses are conducted primarily at the family level.

Table SI.7 shows that family-level performance is strongest for soft propaganda and public service, while the weakest family is state governance, which is also the most semantically intermediate family in the codebook. Table SI.8 shows the same pattern at the ten-label level: performance is strongest for city-image and cultural content, economic development, leadership events, and emergency communication, and weaker for mass mobilization, policy disclosure, welfare content, and general public-service information. Those are also the labels where discovery review and coder adjudication revealed the greatest boundary pressure.

As a complementary face-validity check, Figure SI.1 visualizes family-specific keyword clouds built from the archive's auxiliary keyword field. To keep the clouds informative rather than dominated by generic high-frequency terms, the plotted keywords are weighted by family-level TF-IDF rather than by raw counts alone, after splitting the keyword strings on punctuation delimiters. The resulting semantic neighborhoods line up closely with the intended taxonomy: hard-propaganda titles emphasize political leaders, meetings, and ideological-study language; public-service titles emphasize health, weather, safety, recruitment, and epidemic-response language; state-governance titles emphasize policy release, disclosure, rectification, and participation; and soft-propaganda titles emphasize investment, tourism, culture, projects, and development performance. This figure is not a substitute for the coder audit, but it provides an additional full-corpus check that the higher-order families are semantically coherent.

Figure SI.2 visualizes the family-level confusion matrix. Most remaining error is not a collapse of propaganda and service into one undifferentiated category. Instead, it is concentrated in the expected intermediate zone where state-governance titles can be absorbed into adjacent categories. That pattern is consistent with the codebook logic and helps justify using the four families for

Table SI.8: Class-wise validation metrics for the ten detailed labels.

detailed_label	family	precision	recall	f1	support
Public Service Information	Public Service	0.6911	0.8019	0.7424	212
City Image and Cultural Activity	Soft Propaganda	0.9280	0.8974	0.9125	273
Emergency and Risk Communication	Public Service	0.8000	0.8848	0.8403	217
Ideological Education	Hard Propaganda	0.7304	0.9313	0.8187	160
Policy and Administrative Disclosure	State Governance	0.6875	0.7857	0.7333	168
Leadership and Political Events	Hard Propaganda	0.8750	0.8145	0.8437	275
Social Protection and Welfare	Public Service	0.8649	0.6368	0.7335	201
Social Governance and Enforcement	State Governance	0.8247	0.8081	0.8163	99
Economic Development and Construction	Soft Propaganda	0.9115	0.8035	0.8541	346
Mass Mobilization and Participation	State Governance	0.6667	0.6531	0.6598	49

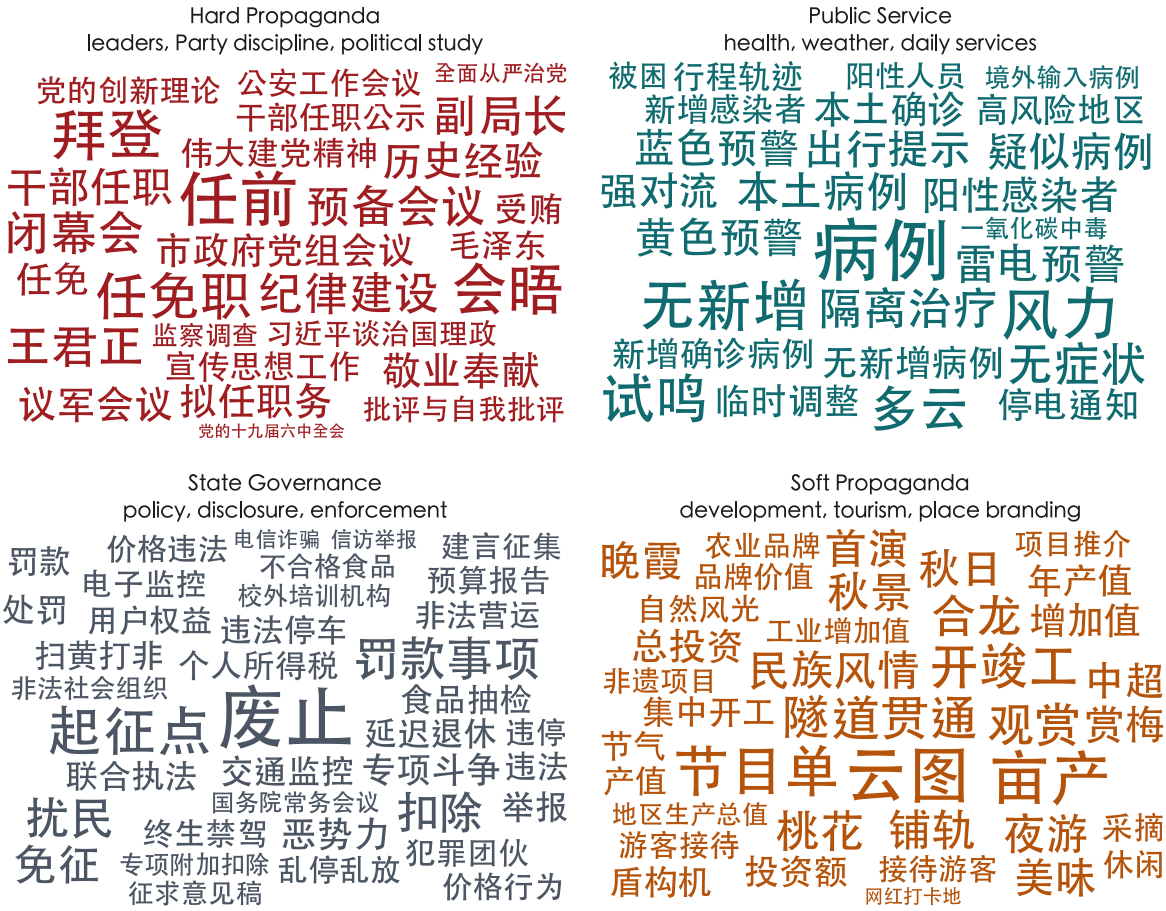

Figure SI.1: Family-Specific Keyword Clouds from the Auxiliary Keyword Field

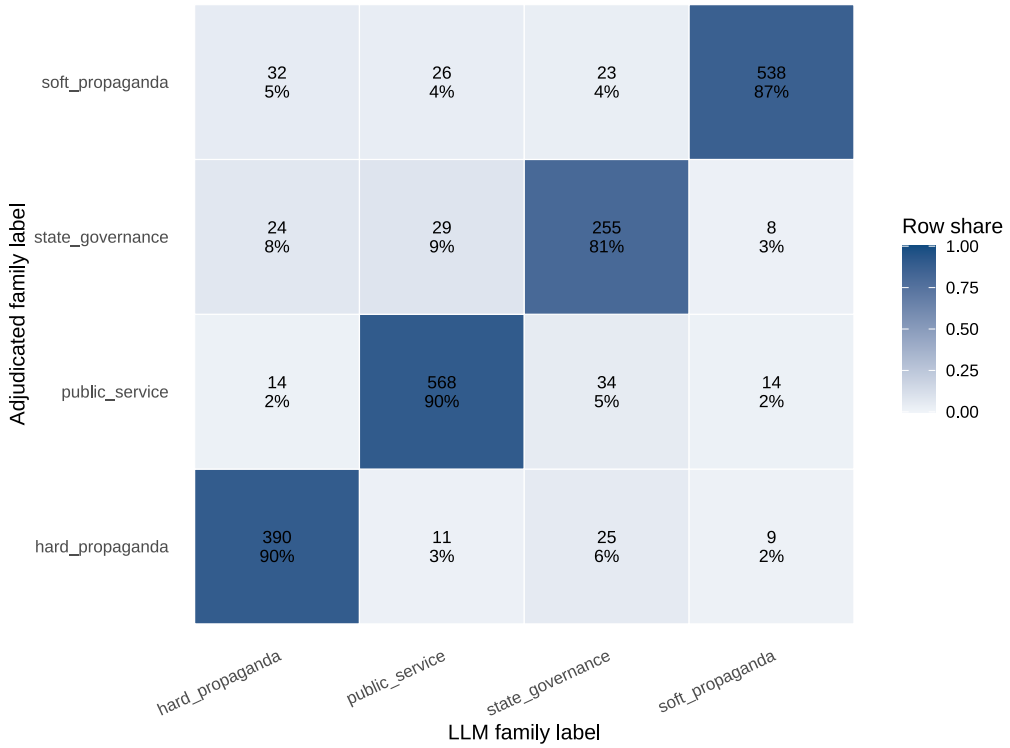

Figure SI.2: Family-Level Confusion Matrix for the WeChat Classification Audit

diagnostics while reserving the three main families for the paper's core theoretical tests.